

Package #110

Design a knee prosthesis with a porous titanium coating combined with a hydrogel-based drug delivery system that promotes bone ingrowth while reducing micromotion at the interface

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'loosening_of_implant_not_related_to_bone' for knee prosthesis: 18 MAUDE reports across 4 distinct devices (20210719 to 20250827); root-cause evidence from 4 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 4 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 10 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, prosthesis, recurring,

Mechanism: Multi-physics hybrid mechanism combining mechanical fixation and bioactive surface treatment

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.